

KHANBOGD, MongoliaWMO: **443850**Lat: **43.1957N**Lon: **107.1940E**Elev: **2999**StdP: **13.17**Time Zone: **8.00 (E08)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.4	-4.9	-22.7	1.8	-0.5	-18.7	2.2	2.7	31.8	14.9	27.8	17.8	6.3	270	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	94.2	63.2	91.2	62.2	88.6	61.1	67.4	86.2	65.3	84.6	63.8	82.6	9.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	91.3	73.4	58.8	82.6	74.0	56.3	75.4	73.9	33.7	87.3	32.0	84.3	30.7	82.9	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
31.1	26.8	22.7	DB	-13.4	99.5	3.9	3.5	-16.2	101.9	-18.5	104.0	-20.7	105.9	-23.5	108.4	
			WB	-13.9	70.1	3.8	3.0	-16.6	72.3	-18.9	74.0	-21.0	75.7	-23.8	77.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	12.0	18.5	34.5	50.2	62.2	72.4	77.9	74.2	63.5	47.7	29.4	15.0
	DBStd	24.78	9.32	10.80	11.29	10.40	9.29	7.08	5.79	6.14	8.15	9.80	10.20	8.63
	HDD50	4559	1177	882	492	123	14	0	0	0	9	158	619	1085
	HDD65	7807	1642	1302	945	450	163	19	1	6	122	539	1068	1550
	CDD50	3324	0	0	12	130	393	673	864	752	412	87	1	0
	CDD65	1098	0	0	0	8	78	241	400	292	76	3	0	0
	CDH74	11213	0	0	0	84	840	2504	4421	2818	527	19	0	0
	CDH80	4383	0	0	0	14	258	961	1981	1075	92	2	0	0
Wind	WSAvg	9.8	9.3	9.5	10.7	11.5	12.1	9.3	8.6	7.8	8.5	9.5	10.4	10.0
Precipitation	PrecAvg	4.3	0.1	0.0	0.1	0.2	0.4	0.8	1.1	0.9	0.4	0.2	0.1	0.0
	PrecMax	6.9	0.2	0.3	0.8	0.6	2.9	2.2	3.0	3.1	2.0	0.6	0.4	0.3
	PrecMin	2.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.1	0.0	0.1	0.2	0.2	0.7	0.6	0.7	0.8	0.4	0.2	0.1	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	48.4	67.1	82.2	91.3	96.1	99.6	96.1	85.9	76.4	57.6	38.8
		MCWB	29.5	35.5	44.3	53.0	56.8	62.1	66.2	64.4	57.5	51.8	41.4	28.5
	2%	DB	33.2	43.4	60.9	75.8	86.3	92.0	95.3	92.1	82.6	71.0	52.2	33.7
		MCWB	25.4	32.0	41.2	49.4	54.9	60.7	64.0	63.8	56.8	49.5	38.2	25.8
	5%	DB	28.5	38.8	56.5	71.7	82.3	88.7	91.9	89.0	79.9	66.4	47.6	30.2
		MCWB	22.0	29.1	38.9	47.9	53.3	59.3	63.5	61.9	56.0	47.0	35.5	23.3
	10%	DB	24.8	34.5	51.6	67.2	78.6	85.5	89.4	85.9	76.9	62.4	43.9	27.0
		MCWB	19.3	25.9	36.3	45.3	51.8	58.3	62.5	61.2	54.7	44.8	33.0	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.7	36.2	44.7	53.6	58.4	65.6	70.3	70.5	62.8	53.9	42.1	29.1
		MCDB	38.3	46.0	64.7	78.7	84.7	84.5	91.2	81.2	76.3	71.7	57.0	37.9
	2%	WB	25.7	32.2	41.8	50.3	56.4	63.5	67.9	67.3	60.0	50.6	38.6	26.0
		MCDB	32.8	43.2	60.2	72.8	82.0	84.4	87.2	84.8	76.8	69.5	51.8	33.3
	5%	WB	22.1	29.2	39.1	48.2	54.6	61.9	65.9	65.2	58.1	47.7	36.0	23.6
		MCDB	28.3	38.8	56.0	71.0	79.6	82.6	86.3	83.5	75.5	65.0	47.6	30.0
	10%	WB	19.4	26.1	36.5	45.9	52.7	60.5	64.4	63.3	56.2	45.4	33.3	21.2
		MCDB	24.7	34.3	51.2	66.6	76.5	81.2	83.5	81.2	73.4	61.6	44.1	26.9
Mean Daily Temperature Range	5% DB	MDBR	15.6	18.8	20.6	20.7	21.1	19.2	18.4	18.3	18.4	18.3	16.1	14.2
		MCDBR	18.0	22.1	23.8	24.5	23.7	21.4	19.9	20.3	20.0	20.7	18.7	16.3
		MCWBR	13.3	15.4	14.1	13.3	11.3	9.1	7.5	8.2	9.2	11.5	11.7	11.8
	5% WB	MCDBR	17.9	21.5	23.5	23.6	22.8	19.8	18.4	17.9	18.1	20.7	18.6	16.1
		MCWBR	13.4	15.1	14.2	13.3	12.1	8.9	7.9	8.5	9.4	12.5	11.9	11.8
Clear-Sky Solar Irradiance	taub	0.283	0.315	0.360	0.411	0.447	0.471	0.447	0.426	0.386	0.340	0.299	0.281	
	taud	2.275	2.201	2.095	2.004	1.959	1.984	2.120	2.154	2.202	2.262	2.297	2.270	
	Ebn at Noon	268	276	276	270	263	256	261	263	265	264	259	256	
	Edh at Noon	27	35	44	53	57	56	48	45	40	33	26	25	
All-Sky Solar Radiation	RadAvg	747	1096	1572	1964	2226	2175	2077	1930	1630	1249	837	643	
	RadStd	60	67	94	79	100	101	95	76	63	38	47	54	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page